Beyond Average Prices: Large Computing Loads and Local Electricity-Price Risk

Bitcoin-Market-Adjusted Evidence from Cryptocurrency-Mining Case Studies

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Abstract

AI-driven growth in data-center demand raises questions about how concentrated computing loads affect local electricity markets. Cryptocurrency mining provides a related setting for examining these questions, although mining and AI workloads need not have the same utilization or flexibility. This paper uses separate interrupted-time-series analyses of public price proxies for Massena in NYISO, a regional SOUTH zone–hub comparison associated with the Rockdale research configuration in ERCOT, and two Susquehanna-area candidates in PJM. The primary models include NOAA temperature, Henry Hub gas prices, lagged log mining hashprice, and its post-event interaction. Bitcoin-price and no-Bitcoin specifications are reported separately. None of the seven immediate mean-basis coefficients survives the reported 5-percent adjustment across 42 primary immediate tests. The fitted geometric daily residual second moment increases by 333.2 percent in ERCOT and decreases by 36.7 and 47.0 percent at the two PJM candidates. The ERCOT and second PJM results survive that adjustment; the first PJM result does not. Arithmetic residual-moment ratios still indicate declines at both PJM candidates, while the ERCOT total ratio remains imprecise. Below-fitted-mean variation accounts for 84.8 percent of the ERCOT arithmetic increase and 95.8–96.9 percent of the PJM declines. Directional log results and zero-moment frequencies require separate interpretation. These findings concern mining-related price patterns relevant to the broader integration of large computing loads, not measured AI effects. Nominal event dates, unverified electrical mappings, uneven weather coverage, and absent facility-load measurements prevent causal attribution and direct extrapolation to AI facilities.

Keywords: cryptocurrency mining; large computing loads; AI-adjacent electricity demand; price basis; hashprice; residual volatility; tail risk

JEL codes: L94; Q41; C22

1 Introduction and Context

1.1 AI-driven demand and a related empirical setting

One of the problems in evaluating the electricity requirements of large computing facilities is that total energy use does not explain what happens to prices where that demand is concentrated. The growth of AI-capable infrastructure makes this question particularly relevant. The 2024 Lawrence Berkeley National Laboratory report estimates U.S. data-center electricity consumption at 176 TWh in 2023, approximately 4.4 percent of national electricity use (Shehabi et al., 2024). The International Energy Agency's 2025 assessment identifies AI-related services as an important source of investment in larger data centers and emphasizes the challenges their geographic concentration presents to local grids (International Energy Agency, 2025). These sources motivate the electricity-market question; they are not observations from the sample analyzed here.

Cryptocurrency-mining facilities are adjacent to this increase in computing demand in an economic and electrical sense, even when the facilities themselves do not run AI workloads. They bring concentrated computing requirements into electricity systems and can create questions about local supply, transmission, and demand response that also arise with AI infrastructure. The U.S. Department of Energy discusses sustained AI-related computing and flexible Bitcoin mining within the same broader problem of supplying large computing loads (U.S. Department of Energy, 2026). This connection makes mining a useful related setting, rather than evidence that mining and AI are interchangeable categories.

The relevant similarity is the pattern of net electricity withdrawal and its location in the network. As a conceptual comparison, two facilities drawing the same power at the same connection and times, with the same surrounding system conditions, would present the same demand to a dispatch model regardless of the computing task. The empirical difficulty is establishing whether those conditions are similar. A high annual load factor can conceal very different behavior during the few hours when the system is constrained. Continuous service availability, computing utilization, and electricity demand relative to installed or contracted capacity are also different measures. The supplied data do not establish that the facilities in this study, or AI facilities generally, operate at nearly 100 percent of electrical capacity.

Operational flexibility is therefore part of the comparison, not a distinction that can be assumed away. Riot reported curtailing 11,717 MWh in July 2022 and receiving approximately $9.5 million in power credits (Riot Blockchain, 2022). Google separately reported reducing machine-learning-related power demand during grid events and developing demand-response arrangements for those workloads, while noting that flexibility is limited and location-specific (Terrell, 2025). These are external operator reports, not mechanisms estimated by this paper. They show why it would be inappropriate either to treat every mining facility as continuously inflexible or to assume that AI workloads can never respond to grid conditions.

The paper consequently retains AI-driven demand as part of its motivation and implications while identifying cryptocurrency mining as its empirical setting. Its relevance to AI is the common

problem of integrating large computing loads and interpreting local-price risk. It does not estimate an AI treatment effect, establish a mining-to-AI conversion factor, or bound the effect of an otherwise similar AI facility. Those stronger conclusions would require facility-level demand profiles and a defensible comparison of operating arrangements.

1.2 Research question and contribution

To investigate this problem, the paper examines the Massena/NORTH comparison in NYISO, the inherited SOUTH load-zone/hub comparison associated with the Rockdale research case in ERCOT, and two name-screened Susquehanna-area candidate prices relative to PPL in PJM. The central empirical question is:

> *How do local-to-benchmark mean prices, residual second moments, directional variation, and extreme basis probabilities change around the nominal mining-related events, after accounting for weather, gas prices, and Bitcoin-market conditions?*

It is important to distinguish a nominal event date from a verified commissioning date or measured change in load. The event registry supplies research conventions, while the pricing points are public proxies. In particular, the ERCOT result is a regional zone–hub diagnostic, not a verified Rockdale nodal effect. These limits cannot be resolved merely by increasing the number of price observations.

The main outcome is the local price minus a contemporaneous benchmark in the same market. Mean SOUTH load-zone prices rise from \$32.84 to \$78.02/MWh in the retained raw sample, while SOUTH hub prices rise from \$30.11 to \$76.49/MWh. The raw basis therefore declines. This illustrates why a large absolute-price increase is not sufficient evidence of a location-specific effect. It also does not establish that computing demand had no regional effect (LMP Research Project, 2026e).

Three findings organize the results. First, immediate mean-basis estimates remain unresolved under the reported primary false-discovery-rate adjustment. Second, the hashprice-adjusted log-moment model shows a large positive ERCOT break and negative estimates at both PJM candidates, although only the second PJM result survives that adjustment. Third, arithmetic residual decompositions place most of the ERCOT increase and PJM declines below the fitted mean. The paper reports these alongside the directional log coefficients, tail probabilities, and specification comparisons rather than treating them as interchangeable measures of volatility.

The contribution is to distinguish absolute prices from relative prices, mean changes from distributional changes, and whole-period arithmetic comparisons from conditional event-date estimates. The updated analysis also separates the Bitcoin-price and mining-hashprice specifications from models that omit them. It does not claim that adding Bitcoin controls turns a nominal event into an exogenous change in facility electricity demand.

2 Research Questions and Economic Framework

2.1 Outcomes and hypotheses

The tests consider an immediate mean-basis change, a subsequent trend change, changes in total and directional residual moments, changes in fixed-threshold tail probabilities, and a change in the relationship between hashprice and the outcome after the nominal event. The hypotheses are two-sided. A concentrated load may coincide with tighter local supply, different dispatch, operating flexibility, or supporting infrastructure. No common sign is imposed across the three settings.

For price point i and hour h, define

$$B_{ih} = P_{ih} - P_{b(i)h}, \tag{1}$$

where $b(i)$ is the benchmark in the same market. NYISO and PJM provide locational wholesale prices, while the ERCOT comparison uses load-zone and hub settlement-point prices. An increase in a negative basis toward zero means that the local discount has narrowed, not necessarily that the absolute price separation has grown.

Subtracting the benchmark removes an identical additive movement shared by both prices. It does not create an untreated facility-level counterfactual: a common shock can affect the two series differently, and a regional response to computing demand could affect both and largely disappear from their difference. The basis concerns relative spatial pricing, not the total regional cost of new load. Nor is it exclusively a congestion measure without a separate examination of price components (New York Independent System Operator, 2026; Electric Reliability Council of Texas, 2026; PJM Interconnection, 2026).

2.2 Mining incentives and distributional risk

Network mining revenue per unit of hash capacity provides an external measure of mining-market conditions. The primary specification uses lagged log hashprice rather than only Bitcoin price because the implemented measure combines issuance revenue, fees, price, and network hashrate. This is a revenue indicator, not a measured facility profit or actual load series. An interaction with the post indicator permits its relationship with prices to change after the event; it does not establish that the facility changed its operating behavior in response.

While average basis may change little, the frequency or severity of unusual hours can still change. Positive and negative residuals here refer to deviations from the fitted hourly basis mean, not positive and negative electricity prices or a facility's financial profit and loss. Research on space-time computing flexibility gives reasons to examine price differences and network constraints (Zhang and Zavala, 2021, 2023), but does not identify the operating strategy of the facilities in this sample.

The analysis uses the HAC approach of Newey and West (1987), dependence-preserving block resampling, and reported Benjamini–Hochberg adjustments (Benjamini and Hochberg, 1995). These

methods address uncertainty conditional on a specification. The completed tables are not difference-in-differences treatment effects, synthetic-control estimates, or responses per observed MW. Methods such as Callaway and Sant'Anna (2021) and Rambachan and Roth (2023) become relevant to a stronger design once credible treated and untreated units are assembled.

3 Cases, Event Conventions, and Data Sources

3.1 Research configurations and evidence class

Table 1 records the study windows used in the analysis. Each covers approximately three years before and three years after its nominal event, with the end date excluded. The code retains its earlier Tier C label for mining or legacy computing; here these cases form the mining sample rather than a presumed AI cohort. It is important to distinguish the number of price comparisons from the number of events: the results contain three event conventions and seven non-alias pricing comparisons, not seven independent facility openings. In particular, both PJM candidates share the same event and benchmark.

Table 1: Implemented research configurations

Case / market	Window, end exclusive	Nominal event	Price comparison and qualification
Massena / NYISO	2015-05-01 to 2021-05-01	2018-05-01	Four non-alias public price proxies minus NORTH; primary label `SITHE___MASSENA`. No verified facility meter.
Rockdale / ERCOT	2017-05-01 to 2023-05-01	2020-05-01	`LZ_SOUTH` minus `HB_SOUTH`. SOUTH's electrical relevance to Rockdale remains unverified; interpret as a regional zone–hub comparison.
Nautilus / PJM	2020-02-01 to 2026-02-01	2023-02-01	Two metadata-screened candidate pnodes minus PPL; full source identities and facility connection are not established in the supplied result export.

Source: supplied retrieval and analysis configurations (LMP Research Project, 2026c,d). Alternative coded dates are 2021-05-26 for the ERCOT research case and 2023-04-01 for PJM. These are sensitivity conventions, not new documentary verification. Later AI-related announcements, transactions, or ownership phases are not identified by the earlier PJM event.

The four non-alias Massena labels are `SITHE___MASSENA`, `ALCOA___DSASP`, `HQ_GEN_CEDARS`, and `STLAWRENCE____`. The downloader also obtains `ALCOA_RYNLDS___DRP`, but its observed prices are an exact alias of another series. It is therefore excluded rather than counted as an independent robustness node. The selection of nodes does not depend on which one produces a desired coefficient.

The PJM names used in the results, `SUSQUEHANNA_PNODE_1` and `SUSQUEHANNA_PNODE_2`, are internal labels assigned for analysis. To select candidates, the downloader searches pricing-node metadata for Susquehanna-related and associated names, then ranks them using zone, name, history, and type fields. It attempts to exclude aggregate nodes when physical candidates are available. While this provides a screening procedure, it does not validate the electrical connection to the facility. A final replication release must copy the source pnode IDs, exact source names, and successful query

filters from the saved route manifest. Those details were not included in the submitted statistical tables and are not inferred from the internal labels.

3.2 Price data: three distinct retrieval routes

NYISO. The documented default route obtains real-time integrated-hourly location-based marginal prices from public monthly ZIP archives for zones and generator/resource points. The archive base is `https://mis.nyiso.com/public/csv/rtlbmp`. Files follow the naming conventions `YYYYMM01rtlbmp_zone_csv.zip` and `YYYYMM01rtlbmp_gen_csv.zip`, with daily CSV files inside each archive. NORTH provides the benchmark, and the resource labels provide the candidate local series. While the program also supports the hosted `nyiso_lmp_real_time_hourly` dataset, the supplied configuration selects the public route (New York Independent System Operator, 2026; LMP Research Project, 2026c). The resulting series is hourly integrated real-time pricing; day-ahead and real-time products are not mixed together.

ERCOT. Prices are obtained from ERCOT's *Historical RTM Load Zone and Hub Prices*, product NP6-785-ER, Report Type ID 13061. ERCOT describes this product as real-time settlement-point prices (SPPs) for hubs and load zones, available in ZIP/XLSX files (Electric Reliability Council of Texas, 2026). The program reads the annual historical files and retains ordinary `LZ_SOUTH` and `HB_SOUTH` observations, rather than substituting the separately identified energy-weighted load-zone category. To form a complete UTC hourly arithmetic mean, it requires four distinct 15-minute observations. This distinction in the source matters: a load-zone SPP is not the LMP at a private facility's electrical bus. The legacy parser uses the `gridstatus` package, and the documented route does not require the hosted ERCOT data allowance (LMP Research Project, 2026c).

PJM. The default route uses GridStatus.io's hosted `pjm_lmp_real_time_hourly` dataset together with the `pjm_pricing_nodes` metadata dataset. In this case, PJM supplies the underlying market data and GridStatus.io acts as the retrieval intermediary. The analysis compares hourly total real-time prices at the two selected candidates with the PPL benchmark (PJM Interconnection, 2026; GridStatus, 2026; LMP Research Project, 2026c). While the result export establishes that rows were returned and the analysis completed, it does not contain the full API route manifest or data-vintage history. Reproducing this retrieval therefore requires those records and the corresponding provider access. No new direct PJM archive retrieval is claimed to have been independently run for this revision.

All price variables are expressed in nominal dollars per MWh, and the main outcomes use total prices. Although congestion and losses can contribute to the basis, the completed common analysis does not separately estimate the published energy, congestion, and loss components.

3.3 NOAA weather and station selection

Temperature observations are obtained from NOAA NCEI's Global Hourly / Integrated Surface Database through the Access Data Service. ISD combines hourly and synoptic surface observations and includes temperature along with other meteorological fields (NOAA National Centers for Environmental Information, 2026b,a). To construct the regression series, the program requests year-sized chunks and reads `TMP` as signed tenths of a degree Celsius. It discards the coded missing sentinel and observations with quality flags 2, 3, 6, and 7, then takes the median of the remaining observations within each UTC hour. Thus the implemented processing determines the hourly series rather than an assumption about how frequently a station reports. Missing temperatures are not interpolated from adjacent hours (LMP Research Project, 2026c).

Table 2: Configured weather stations and their analytical roles

Case	NOAA station ID	Configured name and role
Massena	`72622394725`	Massena International Airport; primary station.
Rockdale	`72103400351`	Cameron Municipal Airport; primary station.
Rockdale backup	`72376123901`	Caldwell Municipal Airport in the inherited configuration; supplies observations where primary temperature is missing.
Nautilus / PJM	`72513014777`	Wilkes-Barre/Scranton International Airport; primary station.

Source: configured identifiers and names in LMP Research Project (2026c,d). They are reported as configured, not newly geocoded or ranked for distance in this revision. The supplied tables do not report the fraction of ERCOT hours drawn from the backup. Station-history and fallback-share checks remain part of the data audit.

For ERCOT, the primary and backup station series are matched by UTC hour, and observed backup values are used wherever primary temperature is missing. This substitutes an observation from another station; it does not interpolate across time. No comparable backup is configured for PJM. The baseline uses temperature and its square, rather than every meteorological field the downloader can retain. When these controls are converted to daily frequency, the program averages the hourly temperatures and the squared hourly temperatures separately. This is important because $\overline{T^2}$ is not generally equal to $\overline{T}^2$.

3.4 Fuel-price control

To account for fuel prices, the analysis uses the U.S. Energy Information Administration's daily Henry Hub natural-gas spot price, available through FRED as `DHHNGSP`. The series is measured in dollars per million Btu and is not seasonally adjusted (U.S. Energy Information Administration, 2026). The program retrieves the FRED graph CSV with a lead-in buffer and carries the latest available observation forward for no more than seven calendar days. It does not use future observations to fill earlier gaps. Gas price enters the model directly rather than being multiplied by a heat-rate scenario. As a result, the reported 100-percent gas coverage describes availability *after* this calendar-alignment rule, not a newly reported quote for every day. Henry Hub serves as a common fuel-price indicator; it is not a measured delivered fuel cost for every generator in the three markets.

3.5 Bitcoin-market data and reference values

The repaired analysis obtains the public Coin Metrics Bitcoin CSV and retains its source hash and download timestamp (Coin Metrics, 2026a; LMP Research Project, 2026d). The snapshot was retrieved on September 29, 2026 at 02:58:48 UTC. Its last source date is May 24, 2026, which is later than the latest study end date of February 1, 2026. This is the vintage actually used in the supplied run, not a claim that the archive contains observations through the download day. The code does not forward-fill missing Bitcoin fundamentals.

For source day s, the implemented hashprice is

$$H_s = \frac{I_s^{USD} + F_s^{BTC} P_s^{BTC/USD}}{R_s^{TH/s}/1000}, \tag{2}$$

where I^{USD} is issuance revenue, F^{BTC} is transaction fees in Bitcoin, $P^{BTC/USD}$ is the price, and $R^{TH/s}$ is network hashrate. The source fields are `IssTotUSD`, `FeeTotNtv`, `PriceUSD`, and `HashRate`. Coin Metrics documents Bitcoin hashrate in TH/s (Coin Metrics, 2026b); division by 1,000 expresses the denominator in PH/s. The resulting measure is dollars per PH/s-day. This calculation uses the source issuance-revenue field and values fees at the recorded daily price; it is not a transaction-level reconstruction of miners' realized receipts.

A complete calendar grid is constructed before lagging. Local analysis date d receives the source observation labeled with UTC date $d-1$, rather than the previous available row. The primary regressor is

$$Z_{id} = \log H_{d-1} - c_i, \qquad c_i = \log H_{t_i^*-1} \tag{3}$$

for the event references actually present in this run. The code permits an earlier available reference if necessary, but all three recorded anchors have zero age relative to the nominal event join date. The Bitcoin-price alternative applies the same transformation to BTC/USD. Table 3 records both references. Centering changes the interpretation of the post coefficient, not the information in the regressor.

The one-day rule is a source-date convention, not a point-in-time publication-vintage guarantee. Global network conditions can also respond to energy costs and mining activity; lagging does not make hashprice an instrument or prove exogeneity. No margin that subtracts the focal electricity price enters a price regression. Hardware-efficiency scenarios in the working data are descriptive only. Population, metered facility demand, delivered regional gas prices, and a separate mining-difficulty regressor are not included in the completed common equations.

3.6 Time alignment, validation, and the resulting samples

Prices and weather are matched in UTC; local calendar variables use Eastern time for NYISO/PJM and Central time for ERCOT. Duplicate location–time observations and conflicting revisions are not simply averaged, and incomplete native intervals fail the hourly-price screen. Negative prices and genuine spikes remain in the baseline. Missing prices are not interpolated.

Table 3: Recorded Bitcoin reference conditions

Case	Source UTC date	BTC/USD	Hashprice
Massena	2018-04-30	9,228.73	587.90
ERCOT research case	2020-04-30	8,652.41	145.11
PJM research case	2023-01-31	23,130.05	75.86

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). All three local join dates equal the nominal event dates; the source observations are dated one day earlier. Hashprice is dollars per PH/s-day. These reference values define $Z = 0$; they are not period averages.

The raw comparisons use all paired local/benchmark hours. For the common-sample regressions, price basis, temperature, gas, lagged log Bitcoin price, and lagged log hashprice must all be present. At least 80 percent of hours must satisfy the relevant screen on each side. A usable local day requires at least 90 percent of its expected hours, including 23- and 25-hour daylight-saving days, and each regime must retain at least 90 usable days. Daily regressions give each retained day equal weight.

Both lagged Bitcoin series have complete coverage in every declared site window. The common-sample day counts match the original weather/gas counts, and the original- and common-sample weather/gas estimates agree to numerical precision. The new results therefore do not reflect an additional observed loss of days from Bitcoin availability. Equality of counts alone would not prove identical timestamps, but here the complete Bitcoin coverage and the code's common eligibility rule provide the additional basis for that comparison.

Table 4: Data coverage after restoring Bitcoin controls

Case	Price rows	Hours pre/post	Controls (%)	BTC (%)	Days
Massena	315,648	26,304 / 26,304	99.92 / 99.70	100 / 100	2,182
ERCOT SOUTH	105,168	26,304 / 26,280	99.90 / 99.86	100 / 100	2,184
PJM candidates	157,824	26,304 / 26,304	98.90 / 85.28	100 / 100	2,003

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Counts are per pricing comparison except price rows, which include all downloaded locations. Paired-price and aligned-gas coverage is 100 percent. Both lagged Bitcoin-price and hashprice coverage are 100 percent pre/post. Complete-control coverage is consequently limited by weather. The exact NYISO alias is not counted as another result.

PJM remains the principal missing-data concern. Complete-control coverage falls from 98.90 to 85.28 percent, with 1,072 usable pre-event days and 931 post-event days. The raw price comparisons include hours that the controlled analysis omits. The supplied records do not establish whether the omitted weather observations are concentrated in storms, particular seasons, or unusual prices. Complete Bitcoin coverage does not repair that selection problem, and coverage at one station does not establish adequate control for weather throughout a broad benchmark region.

4 Empirical Strategy and Interpretation

4.1 Raw comparisons and five fitted specifications

Raw hourly mean differences and sample-variance ratios summarize the two regimes without the control screen. Seven-day moving-calendar-block resampling is performed separately within each regime for 499 draws, giving descriptive percentile intervals. The variance-ratio reference is one rather than zero. These calculations do not provide a treatment counterfactual.

The controlled analysis estimates five specifications. The original-sample weather/gas model is retained as an omission comparator. Four common-sample models then include weather alone; weather and gas; weather, gas, lagged log BTC/USD and its post interaction; or weather, gas, lagged log hashprice and its post interaction. Even the common-sample weather-only regression uses the common eligibility screen. Thus the incremental comparisons do not change the sample simply because a variable is absent from one equation. Bitcoin price and hashprice are alternative specifications rather than a combined collection of closely related mining-revenue components.

The primary specification is `hashprice_interaction`, as designated in the supplied script and run manifest. This preference follows the earlier mining-economics formulation rather than choosing the specification with the smallest observed p value. It is not a claim that the full research design was prospectively preregistered. The analysis retains all comparison models and reports a wider adjustment across them.

4.2 Adjusted daily mean basis and Bitcoin sensitivity

For each pricing comparison, the primary daily equation is

$$\begin{aligned}\overline{B}_d = \alpha + \beta D_d + \rho\tau_d + \kappa D_d\tau_d + \mu_{m(d)} + \nu_{w(d)} \\ + \gamma_1\overline{T}_d + \gamma_2\overline{T^2}_d + \gamma_3 G_d + \lambda Z_d + \theta D_d Z_d + u_d.\end{aligned} \tag{4}$$

Here D is the nominal post indicator, τ is time in years relative to the event, and $m(d)$ and $w(d)$ identify month of year and weekday. G is the aligned Henry Hub price. Temperature and squared temperature are separately averaged over the same eligible hours used for the outcome. Coefficients are estimated separately for each comparison using ordinary least squares (LMP Research Project, 2026d).

The immediate coefficient β is the fitted event-date change at the reference Bitcoin condition $Z = 0$. The pre-event time slope is ρ and the post-event time slope, holding Z fixed, is $\rho + \kappa$. The pre-event hashprice association is λ, while the post-event association is $\lambda + \theta$. For example, a 10-percent increase in hashprice changes the fitted post-event mean basis by $(\lambda + \theta)\log(1.10)$, holding other regressors fixed. That is an association with a network revenue measure, not an estimated facility demand elasticity.

At h years after the event and centered log-hashprice z, the model-implied contrast relative to retaining the pre-event relationship is

$$\Delta(h, z) = \beta + \kappa h + \theta z. \tag{5}$$

Its standard error requires the covariance of the relevant coefficients. The exported horizon table uses $z = 0$ at 3, 6, 12, 24, and 36 months where the calendar span allows it. These are reference-condition point contrasts, not averages over those horizons or predictions along the observed Bitcoin path. The post-sensitivity intervals added in this revision use the supplied covariance matrices, with

$$\mathrm{Var}(\widehat{\lambda} + \widehat{\theta}) = \mathrm{Var}(\widehat{\lambda}) + \mathrm{Var}(\widehat{\theta}) + 2\,\mathrm{Cov}(\widehat{\lambda}, \widehat{\theta}). \tag{6}$$

No model was re-estimated to calculate those linear contrasts.

The raw pre/post mean difference and $\widehat{\beta}$ remain different quantities. The regression adds trends, seasonality, covariates, and a selected daily sample. With the interaction, $\widehat{\beta}$ also has a stated Bitcoin reference. A change between model versions therefore cannot be described as a causal fraction of the price movement explained by Bitcoin conditions.

4.3 Hourly residuals and directional second moments

For each specification, an hourly OLS model includes post, time trend, their interaction, hour-of-day, weekday, month, and that specification's continuous controls. In the Bitcoin models, both the lagged centered log term and its post interaction enter this first stage. The fit uses complete eligible hours across the full pre/post sample, not only pre-event data. For residual $e_h = B_h - \widehat{\mu}_h$, define on each usable day

$$M_d = n_d^{-1} \sum_{h \in d} e_h^2, \tag{7}$$

$$M_d^+ = n_d^{-1} \sum_{h \in d} \max(e_h, 0)^2, \tag{8}$$

$$M_d^- = n_d^{-1} \sum_{h \in d} \min(e_h, 0)^2. \tag{9}$$

The denominator includes all valid residual hours, with opposite-sign residuals contributing zero. Thus $M = M^+ + M^-$. These are squared departures from a fitted hourly basis mean, in $(\$/\mathrm{MWh})^2$, not squared price changes, variance of daily means, or a GARCH variance forecast. They include changes in a day's residual mean as well as within-day spread.

The arithmetic pre/post averages and ratios compare whole regimes of retained days. The contribution of the negative component to a total arithmetic change is

$$\frac{\overline{M}^-_{post} - \overline{M}^-_{pre}}{\overline{M}_{post} - \overline{M}_{pre}}, \tag{10}$$

when the denominator is nonzero. This is an accounting decomposition, not a decomposition of causal mechanisms. Restoring Bitcoin controls changes the fitted residuals, so these quantities are recomputed rather than carried over from the earlier paper.

4.4 Log-moment and tail regressions

Equation 4 is then fitted separately for $\log(M_d + \epsilon)$, $\log(M_d^+ + \epsilon)$, and $\log(M_d^- + \epsilon)$, with $\epsilon = 10^{-8}$. Each equation has its own coefficients. The transformation

$$100\{\exp(\widehat{\beta}_M) - 1\} \tag{11}$$

is a fitted geometric-moment contrast at the nominal event and reference hashprice. It is not automatically the percent change in the arithmetic expected moment, raw hourly variance, or standard deviation. In general, $\exp\{\mathbb{E}[\log(M + \epsilon) \mid X]\}$ differs from $\mathbb{E}[M + \epsilon \mid X]$. A correct conditional-log interpretation also depends on the specified model; exponentiation alone does not establish the true conditional mean.

Large days can dominate arithmetic moments, while zero or tiny directional moments can strongly influence the log model. The supplied zero-day counts are therefore reported with the directional coefficients. They show whether the offset is used, but do not quantify its influence on the event coefficient. Alternative offsets and two-part or level-scale models have not been estimated in this run.

For tail outcomes, lower and upper cutoffs are fixed at the 1st and 99th percentiles of all paired pre-event raw basis observations. The cutoffs do not change with Bitcoin eligibility or get recalculated after the event. Hourly exceedance indicators are averaged over eligible hours on usable days, then fitted with the same daily regressors. These are linear probability regressions on daily fractions, not logistic models. Coefficients multiplied by 100 are percentage-point changes. Tail frequency refers to crossing a raw-basis cutoff rather than the sign or size of a fitted residual.

4.5 Uncertainty and multiple comparisons

Daily regressions use a Bartlett HAC covariance with 30 actual calendar days of dependence and a finite-sample degrees-of-freedom correction. Missing dates retain their calendar spacing. Marginal intervals and p values use large-sample normal approximations (Newey and West, 1987). Required Bitcoin terms are checked in the actual design matrix, and rank-deficient designs are rejected; a successful check is not proof that all confounding has been removed.

Benjamini–Hochberg adjustments are reported for the six immediate post outcomes within a node, and across all seven comparisons within a specification. This gives 42 tests per specification; the primary adjusted value is denoted q_{42}. A second adjustment, q_{210}, includes all five specifications and 210 immediate tests (LMP Research Project, 2026e; Benjamini and Hochberg, 1995). These families are reported as implemented, including overlapping specifications and outcomes. They do not establish independence among the comparisons or a guarantee under arbitrary cross-test

dependence. Neither q is the probability that a particular finding is false, and marginal 95-percent intervals are not simultaneous intervals.

The immediate-post adjustments do not apply to time-slope changes, Bitcoin coefficients, interactions, horizon contrasts, event bins, placebo tests, or sensitivity searches. Those are explicitly secondary and unadjusted. An across-specification BH value need not be numerically greater than the within-specification value because the ranked family changes; this is not a reason to choose whichever value is smaller.

Block intervals use 499 draws of seven-day moving blocks from a calendar grid separately within each regime. For residual moments they condition on the already-fitted hourly model, without refitting that first stage or re-estimating thresholds. They are descriptive percentile intervals, not null-centered treatment tests. Further inference would require more draws, alternative dependence blocks, and propagation of first-stage and model-selection uncertainty.

5 Results: Mean Prices and Local Basis

5.1 Raw prices and relative movements

The raw price summaries do not depend on Bitcoin controls. At the primary Massena point, mean local price falls from \$19.53 to \$17.09/MWh, while NORTH falls from \$19.79 to \$16.88. The signed basis nevertheless rises from about −\$0.267 to \$0.218/MWh. In ERCOT, the local-price increase of \$45.19/MWh is slightly smaller than the hub increase of \$46.38, leaving a raw basis change of −\$1.20/MWh. Neither comparison by itself identifies a causal regional response.

For PJM, mean local prices fall from \$36.32 to \$29.23/MWh and from \$36.75 to \$29.80/MWh, while PPL falls from \$41.24 to \$32.04. Their raw bases rise by \$2.10 and \$2.25 because the local discounts narrow. This is not a raw increase in local LMP. Table 5 retains the paired-hour descriptive comparisons from the new export.

Table 5: Unadjusted hourly basis means before and after the nominal event

Comparison	Pre mean	Post mean	Difference	95% block CI
Massena: SITHE	-0.267	0.218	0.485	[0.234, 0.788]
Massena: ALCOA	-0.140	-0.227	-0.087	[-0.165, -0.006]
Massena: Cedars	-0.355	0.062	0.416	[0.112, 0.771]
Massena: St Lawrence	-0.334	-0.364	-0.030	[-0.135, 0.062]
ERCOT: SOUTH	2.726	1.529	-1.197	[-2.933, 0.714]
PJM: candidate 1	-4.916	-2.815	2.101	[1.129, 3.332]
PJM: candidate 2	-4.489	-2.243	2.247	[1.202, 3.472]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Units: dollars per MWh. These are paired-hour period averages without a weather, gas, or Bitcoin sample screen. Intervals use 499 seven-day block draws and are descriptive. The raw tables are retained, not reinterpreted as event-date regression coefficients.

The raw basis intervals exclude zero for SITHE and Cedars on the positive side and for ALCOA

on the negative side. The ERCOT interval includes zero, while both PJM intervals exclude it on the positive side. These are statements about the raw period averages, not substitutes for the adjusted coefficients.

5.2 Primary immediate mean-basis changes

Table 6 reports the primary hashprice-adjusted immediate coefficients. None of the seven passes the reported 5-percent q_{42} or q_{210} threshold. This does not demonstrate an absence of economically important changes: the primary SITHE interval extends to approximately \$1.97/MWh, while ERCOT's upper endpoint is approximately \$10.89/MWh. The results are unresolved rather than estimates of exactly zero.

Table 6: Immediate changes in daily mean basis: primary hashprice specification

Pricing comparison	$\widehat{\beta}$	95% CI	p	q_{42}	q_{210}
Massena: SITHE	0.943	[-0.079, 1.966]	0.0706	0.1407	0.1561
Massena: ALCOA	-0.075	[-0.311, 0.160]	0.5318	0.6382	0.6290
Massena: Cedars	0.967	[-0.092, 2.026]	0.0735	0.1407	0.1596
Massena: St Lawrence	0.098	[-0.149, 0.346]	0.4354	0.5379	0.5291
ERCOT: SOUTH	4.451	[-1.989, 10.890]	0.1755	0.2730	0.2899
PJM: candidate 1	2.659	[-0.209, 5.527]	0.0692	0.1407	0.1546
PJM: candidate 2	3.414	[0.186, 6.641]	0.0382	0.1069	0.1003

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Units: dollars per MWh. Primary hashprice-interaction specification. The immediate contrast holds hashprice at its site-specific event reference. Marginal intervals use calendar-HAC covariance. q_{42} adjusts immediate post tests within this specification; q_{210} includes all five specifications.

For Massena, the SITHE and Cedars coefficients are approximately \$0.943 and \$0.967/MWh, with raw p values of 0.0706 and 0.0735. ALCOA and St Lawrence remain closer to zero. ERCOT's estimate is \$4.451/MWh, with interval $[-1.989, 10.890]$ and $p = 0.1755$. Its positive conditional break and negative raw period difference are not contradictory estimates of the same quantity: they use different weighting and a different comparison, with the event coefficient evaluated at the recorded hashprice reference.

PJM candidate 1 has an immediate estimate of \$2.659/MWh with $p = 0.0692$. Candidate 2 has an estimate of \$3.414/MWh and a marginal interval excluding zero, with $p = 0.0382$. Its $q_{42} = 0.1069$ and $q_{210} = 0.1003$, however, do not pass 0.05. The evidence therefore does not establish a statistically robust facility-induced increase in local electricity prices.

The updated slope table also requires care. Cedars has a nominal mean-basis slope-change $p = 0.0494$, unlike the earlier no-Bitcoin version in which no mean-slope test crossed 0.05. That is an unadjusted secondary result with an interval only just above zero, not a confirmed persistent treatment effect. Appendix A reports all mean and total-moment trend changes.

6 Results: Total Residual Variation

6.1 Immediate fitted geometric moments

The primary log-second-moment results differ from the mean results. None of the immediate Massena total-moment coefficients rejects zero even before multiplicity adjustment. The transformed primary SITHE estimate is +6.4 percent, with a wide marginal interval from −53.8 to +145.4 percent. The other Massena intervals are also broad.

Table 7: Immediate changes in the fitted geometric daily residual second moment

Pricing comparison	Change (%)	95% CI (%)	p	q_{42}	q_{210}
Massena: SITHE	6.4	[-53.8, 145.4]	0.8839	0.9055	0.8924
Massena: ALCOA	1.3	[-50.8, 108.9]	0.9712	0.9712	0.9721
Massena: Cedars	47.7	[-39.6, 261.4]	0.3927	0.5320	0.4967
Massena: St Lawrence	81.0	[-15.7, 289.0]	0.1282	0.2244	0.2329
ERCOT: SOUTH	333.2	[181.0, 567.9]	< 0.001	< 0.001	< 0.001
PJM: candidate 1	-36.7	[-57.2, -6.5]	0.0215	0.0751	0.0601
PJM: candidate 2	-47.0	[-65.2, -19.1]	0.0032	0.0122	0.0105

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Transformation: $100[\exp(\widehat{\beta}_M) - 1]$ for $\log(M_d + 10^{-8})$. This is a fitted geometric-moment contrast, not an arithmetic variance or standard-deviation change. Primary hashprice-interaction specification. The immediate contrast holds hashprice at its site-specific event reference. Marginal intervals use calendar-HAC covariance. q_{42} adjusts immediate post tests within this specification; q_{210} includes all five specifications.

ERCOT's primary log coefficient is 1.466045, corresponding to a 333.2-percent fitted geometric-moment increase at the event and reference hashprice. Its transformed marginal interval is $[181.0, 567.9]$ percent. The raw p value is approximately 3.19×10^{-11}, with $q_{42} \approx 4.46 \times 10^{-10}$. This is strong evidence of a break within the specified log-moment model, not evidence that ordinary hourly variance increased by exactly 333 percent or that the mining facility caused that increase.

The two PJM coefficients remain negative, but they no longer have the same inference status. Candidate 1's log coefficient is −0.457836, a fitted geometric-moment change of −36.7 percent with marginal interval $[-57.2, -6.5]$. Its raw $p = 0.0215$ excludes zero at the nominal level, but $q_{42} = 0.0751$ and $q_{210} = 0.0601$ do not pass 0.05. Candidate 2's coefficient is −0.634190, corresponding to −47.0 percent with interval $[-65.2, -19.1]$; its $q_{42} = 0.0122$ and $q_{210} = 0.0105$ do pass. The old statement that both PJM total-moment coefficients survive the primary 5-percent correction cannot be retained.

6.2 Arithmetic moments and raw variance

The arithmetic total-moment comparisons remain similar in direction to the earlier calculations, but they are recomputed from the hashprice-adjusted residuals. ERCOT's average daily total moment rises from 732.29 to 1,363.13, a ratio of 1.861 with descriptive interval $[0.591, 4.513]$. Its raw hourly basis-variance ratio is 1.879 with interval $[0.641, 4.702]$. Both intervals include one. The precision

of the log-regression break therefore does not extend to every measure of whole-period variance change.

PJM's arithmetic residual-moment ratios are 0.174 and 0.175, with descriptive intervals $[0.097, 0.300]$ and $[0.091, 0.339]$, respectively. Both are below one, corresponding to reductions of approximately 82.6 and 82.5 percent in these arithmetic averages. The raw hourly basis-variance ratios, 0.254 and 0.293, also have intervals below one. These summaries remain informative even though the first candidate's immediate log-regression coefficient does not pass FDR. The quantities differ because of centering, daily weighting, covariate adjustment, trends, and the logarithmic transformation. They do not establish a common numerical effect or a causal stabilization of the grid.

6.3 What changes when the controls change

Table 8 shows the same-sample specifications for ERCOT and PJM; the corresponding Massena results are in Appendix B.1. Adding gas changes the PJM estimates substantially: the weather-only immediate mean coefficients are \$6.790 and \$6.239/MWh, compared with \$2.954 and \$3.653 in the weather/gas model. Adding hashprice and its interaction gives \$2.659 and \$3.414. These comparisons demonstrate specification sensitivity, not a causal allocation of the price movement to gas and Bitcoin.

Table 8: ERCOT and PJM: common-sample control specifications

Comparison	Controls	Mean $\widehat{\beta}$	Mean p	Moment (%)	Moment q_{42}
ERCOT: SOUTH	Weather	4.625	0.1035	219.3	< 0.001
ERCOT: SOUTH	Weather + gas	4.168	0.1275	265.8	< 0.001
ERCOT: SOUTH	BTC-price interaction	5.817	0.1537	745.6	< 0.001
ERCOT: SOUTH	Hashprice interaction	4.451	0.1755	333.2	< 0.001
PJM: candidate 1	Weather	6.790	< 0.001	-78.8	< 0.001
PJM: candidate 1	Weather + gas	2.954	0.0546	-51.6	< 0.001
PJM: candidate 1	BTC-price interaction	2.726	0.0766	-42.7	0.0173
PJM: candidate 1	Hashprice interaction	2.659	0.0692	-36.7	0.0751
PJM: candidate 2	Weather	6.239	< 0.001	-77.0	< 0.001
PJM: candidate 2	Weather + gas	3.653	0.0311	-58.5	< 0.001
PJM: candidate 2	BTC-price interaction	3.539	0.0436	-55.5	< 0.001
PJM: candidate 2	Hashprice interaction	3.414	0.0382	-47.0	0.0122

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). All four specifications use the common eligible sample. Mean coefficients are dollars per MWh; moment entries are $100[\exp(\widehat{\beta}_M) - 1]$, not raw variance changes. q_{42} is recomputed within each specification. The original-sample weather/gas model agrees with its common-sample counterpart and is retained in the accompanying CSVs. Bitcoin-price and hashprice models are alternatives, not simultaneous regressors.

Relative to the no-Bitcoin model, the primary ERCOT fitted geometric-moment increase rises from 265.8 to 333.2 percent. The two PJM decreases become smaller, from 51.6 and 58.5 percent to 36.7 and 47.0 percent. The separate Bitcoin-price specification gives yet another set of estimates.

In particular, the first PJM total-moment result passes its within-specification adjustment in the Bitcoin-price model but not the primary hashprice model. This is a reason to state the selected specification clearly, not to replace the primary result with the more favorable alternative.

The old weather/gas model is now a comparison specification, not the final Bitcoin-adjusted model. All five fitted versions remain in the accompanying numerical files. Since Bitcoin coverage is complete and the weather/gas versions agree, the changes in these fits are associated with the changed specification rather than a recorded loss of sample days.

7 Directional Composition and Tail Behavior

7.1 Arithmetic residual moments in ERCOT and PJM

The identity $M = M^+ + M^-$ shows where the arithmetic changes occur relative to the fitted mean. ERCOT's negative component rises from 23.33 to 558.56, while its positive component rises from 708.96 to 804.58. The negative-component ratio is 23.939, with a descriptive interval entirely above one; the positive-component interval includes one.

Table 9: ERCOT and PJM: primary arithmetic daily residual moments by direction

Comparison	Moment	Pre level	Post level	Ratio	95% block CI
ERCOT: SOUTH	Total M	732.29	1,363.13	1.861	[0.591, 4.513]
ERCOT: SOUTH	Positive M^+	708.96	804.58	1.135	[0.288, 2.948]
ERCOT: SOUTH	Negative M^-	23.33	558.56	23.939	[5.459, 52.320]
PJM: candidate 1	Total M	316.72	55.04	0.174	[0.097, 0.300]
PJM: candidate 1	Positive M^+	23.32	12.36	0.530	[0.378, 0.709]
PJM: candidate 1	Negative M^-	293.40	42.68	0.145	[0.071, 0.273]
PJM: candidate 2	Total M	323.12	56.48	0.175	[0.091, 0.339]
PJM: candidate 2	Positive M^+	20.05	11.79	0.588	[0.441, 0.759]
PJM: candidate 2	Negative M^-	303.07	44.69	0.147	[0.067, 0.318]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice-adjusted hourly residuals. Levels are equal-weighted arithmetic means of usable daily moments, in $(\$/\text{MWh})^2$. All sign-specific moments use the same denominator. Seven-day, 499-draw percentile intervals condition on the fitted first stage and are descriptive.

The negative component accounts for 84.8 percent of ERCOT's total arithmetic increase. Its share of total residual variation rises from 3.2 to 41.0 percent. These are revised figures from the Bitcoin-adjusted residuals, not the old decomposition rounded differently. The interpretation is greater squared basis variation below the fitted hourly mean; it does not imply that absolute electricity prices were negative or that curtailment occurred in those hours.

For PJM, the negative component falls from 293.40 to 42.68 at candidate 1 and from 303.07 to 44.69 at candidate 2. The positive components also fall, from 23.32 to 12.36 and from 20.05 to 11.79. Both positive- and negative-component ratio intervals are below one. Negative-component

reductions account for 95.8 and 96.9 percent of the total arithmetic declines. The negative shares remain large after the event, at approximately 77.5 and 79.1 percent, but their levels are lower.

The raw basis distributions also retain a pronounced lower-side change. Candidate 1's first percentile moves from about −$81.45 to −$36.25/MWh, and candidate 2's from −$78.43 to −$35.62. Their 99th percentiles rise from $2.71 to $6.63 and from $3.19 to $7.00. A decline in total dispersion therefore need not mean that every part of the distribution contracts. These raw quantiles do not receive the regression's Bitcoin adjustment.

7.2 Log-semivariance and the observed zero-day frequencies

The conditional directional log coefficients still do not align uniformly with arithmetic period changes. ERCOT's immediate positive-component coefficient is −8.362 although its arithmetic positive moment rises slightly. PJM's immediate negative-component coefficients are positive and imprecise despite the large declines in their arithmetic negative moments. Table 10 reports these results without treating disagreement as either a coding error or an identified mechanism.

Table 10: Primary immediate log-semivariance coefficients

Comparison	Moment	$\widehat{\beta}$	95% CI	p	q_{42}
Massena: SITHE	M^+	-5.454	[-6.944, -3.964]	< 0.001	< 0.001
Massena: SITHE	M^-	5.693	[4.022, 7.363]	< 0.001	< 0.001
Massena: ALCOA	M^+	1.359	[-0.329, 3.047]	0.1147	0.2094
Massena: ALCOA	M^-	-0.461	[-1.574, 0.653]	0.4175	0.5379
Massena: Cedars	M^+	-5.945	[-7.791, -4.098]	< 0.001	< 0.001
Massena: Cedars	M^-	4.666	[3.187, 6.146]	< 0.001	< 0.001
Massena: St Lawrence	M^+	-0.644	[-2.228, 0.940]	0.4257	0.5379
Massena: St Lawrence	M^-	1.812	[-0.120, 3.744]	0.0660	0.1407
ERCOT: SOUTH	M^+	-8.362	[-11.138, -5.585]	< 0.001	< 0.001
ERCOT: SOUTH	M^-	1.891	[1.259, 2.522]	< 0.001	< 0.001
PJM: candidate 1	M^+	-2.257	[-3.559, -0.955]	< 0.001	0.0028
PJM: candidate 1	M^-	2.633	[-1.593, 6.859]	0.2220	0.3265
PJM: candidate 2	M^+	-2.595	[-4.053, -1.138]	< 0.001	0.0022
PJM: candidate 2	M^-	2.785	[-1.718, 7.289]	0.2254	0.3265

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Coefficients are in log units for $\log(M^{\pm} + 10^{-8})$ at the event/reference hashprice. They are not arithmetic-period ratios. Zero-day frequencies are reported in Table 11; alternative-offset estimates are not supplied.

Unlike the earlier export, the new package includes zero-moment diagnostics. ERCOT's positive daily moment equals zero on 27.3 percent of usable pre-event days and 22.4 percent of post-event days. PJM's negative moment equals zero on approximately 18.8 and 18.7 percent of pre-event days, falling to 6.1 and 6.4 percent afterward. Consequently, the offset is not merely a numerical detail for those directional models.

Fewer negative zero-days in PJM can coexist with a smaller arithmetic negative moment: variation may occur on more days but be less severe on average. The log regression additionally

Table 11: Usable days with a zero directional residual moment

Comparison	M^+ pre (%)	M^+ post (%)	M^- pre (%)	M^- post (%)
Massena: SITHE	3.9	11.9	12.3	2.8
Massena: ALCOA	12.2	3.3	1.9	7.2
Massena: Cedars	5.6	13.4	10.0	4.5
Massena: St Lawrence	4.8	2.8	9.6	9.7
ERCOT: SOUTH	27.3	22.4	0.8	3.0
PJM: candidate 1	1.2	3.7	18.8	6.1
PJM: candidate 2	1.6	3.1	18.7	6.4

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Shares use each node/regime's usable days, not all calendar days. Total daily moments are nonzero in the supplied primary sample. The counts establish the presence of zeros, not the amount of coefficient sensitivity to the chosen offset.

conditions on trends and covariates and evaluates a break at a reference point. The zero shares alone do not establish which of those differences explains its positive coefficient. The main directional accounting statement is therefore tied to the arithmetic moments. Level-scale or two-part regressions, alternative offsets, and a full-refit bootstrap remain uncompleted checks.

Massena also demonstrates why the total moment should not be read as the entire distribution. Its SITHE and Cedars arithmetic positive moments rise with descriptive ratio intervals above one, while ALCOA's negative moment rises. Some directional log coefficients have different signs. The complete primary arithmetic decomposition is retained in Appendix B.

7.3 Fixed-threshold tail probabilities

Table 12 updates all seven comparisons. St Lawrence's upper-tail coefficient is now 6.14 percentage points, with marginal interval $[2.91, 9.38]$ and $q_{42} = 0.0010$. This remains a supported primary upper-tail change despite its unresolved total-moment break. It concerns the probability of exceeding a fixed raw-basis threshold, not a general increase in every variance measure.

ERCOT's upper-tail estimate is 1.82 percentage points with a marginal interval of $[0.23, 3.40]$. Its raw $p = 0.0248$ is below 0.05, but $q_{42} = 0.0801$ and $q_{210} = 0.0685$ are not. The earlier borderline FDR finding no longer holds under the primary hashprice model. PJM candidate 2's upper-tail estimate is 2.62 points with a marginal interval excluding zero, but its $q_{42} = 0.0874$ also exceeds 0.05. Candidate 1 and all lower-tail coefficients remain unresolved after the primary correction. The paper does not select the unadjusted threshold when the adjusted result is less favorable.

Table 12: Primary immediate changes in fixed-threshold basis exceedance probabilities

Comparison	Tail	Change (pp)	95% CI (pp)	p	q_{42}
Massena: SITHE	Lower 1%	-1.69	[-3.42, 0.04]	0.0558	0.1407
Massena: SITHE	Upper 1%	1.34	[-0.48, 3.17]	0.1488	0.2500
Massena: ALCOA	Lower 1%	-0.39	[-2.00, 1.23]	0.6392	0.7065
Massena: ALCOA	Upper 1%	1.47	[-0.64, 3.59]	0.1725	0.2730
Massena: Cedars	Lower 1%	0.51	[-1.47, 2.49]	0.6167	0.7001
Massena: Cedars	Upper 1%	1.60	[-0.15, 3.36]	0.0737	0.1407
Massena: St Lawrence	Lower 1%	-0.33	[-1.94, 1.27]	0.6829	0.7354
Massena: St Lawrence	Upper 1%	6.14	[2.91, 9.38]	< 0.001	0.0010
ERCOT: SOUTH	Lower 1%	-0.45	[-1.97, 1.07]	0.5628	0.6565
ERCOT: SOUTH	Upper 1%	1.82	[0.23, 3.40]	0.0248	0.0801
PJM: candidate 1	Lower 1%	-0.16	[-1.21, 0.90]	0.7735	0.8121
PJM: candidate 1	Upper 1%	2.32	[-0.13, 4.76]	0.0635	0.1407
PJM: candidate 2	Lower 1%	-0.63	[-1.79, 0.52]	0.2828	0.3959
PJM: candidate 2	Upper 1%	2.62	[0.27, 4.97]	0.0291	0.0874

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice specification. Thresholds are fixed at the pre-event 1st and 99th percentiles of paired raw basis. Event coefficients and intervals are multiplied by 100 to give percentage points, evaluated at the reference hashprice. These are frequency measures, not residual-severity measures.

8 Bitcoin Sensitivity and Secondary Dynamics

8.1 Mean-basis sensitivity to hashprice

The interaction coefficient asks whether the relationship between lagged log hashprice and mean basis changes after the event. None of the seven mean-basis hashprice-interaction coefficients rejects zero at the nominal 5-percent level. Their inclusion remains important to the specification, but the results do not identify a common change in mean-basis sensitivity to mining revenue conditions.

The post sensitivities in Table 13 combine the main hashprice coefficient and its interaction using their covariance. Their intervals also include zero. The lack of a precise interaction does not prove that mining economics are irrelevant to actual utilization; the outcome is a public price difference, not electricity consumed by the facility. The adjustment may also be correlated with time trends and other market conditions.

8.2 Total-moment sensitivity to hashprice

The ERCOT log-total-moment relationship is different. Its pre-event log-hashprice coefficient is 0.577, and the change after the event is -1.184 with a marginal interval entirely below zero. The derived post-event slope is -0.607, with interval approximately $[-0.904, -0.310]$. These are unadjusted secondary findings and should not be mistaken for the immediate post-event q_{42} test.

The negative ERCOT post slope is consistent with a fitted association between higher lagged hashprice and a smaller geometric residual moment in the post period. It does not establish a

Table 13: Hashprice sensitivity in the primary daily mean-basis model

Comparison	Pre $\widehat{\lambda}$	Change $\widehat{\theta}$	$p(\theta)$	Post $\widehat{\lambda}+\widehat{\theta}$	95% CI (post)
Massena: SITHE	0.226	0.171	0.5313	0.397	[-0.062, 0.856]
Massena: ALCOA	0.049	-0.106	0.3280	-0.057	[-0.214, 0.100]
Massena: Cedars	0.226	0.293	0.3475	0.519	[-0.006, 1.043]
Massena: St Lawrence	0.053	-0.057	0.6554	-0.004	[-0.166, 0.157]
ERCOT: SOUTH	0.684	-1.886	0.3290	-1.202	[-4.389, 1.985]
PJM: candidate 1	-0.880	-0.894	0.4878	-1.774	[-3.829, 0.282]
PJM: candidate 2	-0.748	-0.858	0.5215	-1.607	[-3.686, 0.473]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Pre and change coefficients come from the primary model. Post sensitivities and their intervals are algebraic contrasts derived for this revision from the supplied coefficients and calendar-HAC covariance matrices; no regressions were rerun. Units: dollars per MWh per log unit of hashprice. All tests here are exploratory and unadjusted; the immediate-post q_{42} does not apply to them.

Table 14: Hashprice sensitivity in the primary log-total-moment model

Comparison	Pre $\widehat{\lambda}$	Change $\widehat{\theta}$	$p(\theta)$	Post $\widehat{\lambda}+\widehat{\theta}$	95% CI (post)
Massena: SITHE	0.002	0.490	0.2061	0.492	[0.012, 0.972]
Massena: ALCOA	0.152	-0.040	0.9003	0.112	[-0.400, 0.624]
Massena: Cedars	-0.044	0.585	0.1400	0.542	[0.011, 1.072]
Massena: St Lawrence	0.206	0.058	0.8517	0.264	[-0.242, 0.770]
ERCOT: SOUTH	0.577	-1.184	< 0.001	-0.607	[-0.904, -0.310]
PJM: candidate 1	0.312	-0.383	0.2029	-0.071	[-0.640, 0.498]
PJM: candidate 2	0.250	-0.484	0.1065	-0.234	[-0.801, 0.333]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Pre and change coefficients come from the primary model. Post sensitivities and their intervals are algebraic contrasts derived for this revision from the supplied coefficients and calendar-HAC covariance matrices; no regressions were rerun. Log–log slopes for the offset residual moment. All tests here are exploratory and unadjusted; the immediate-post q_{42} does not apply to them.

demand-response mechanism or a causal effect of hashprice. Both PJM changes in total-moment sensitivity are imprecise, and their derived post slopes have intervals including zero. SITHE and Cedars have nominally positive derived post slopes, but their changes in slope are not individually significant. A significant post-period slope and an insignificant change in that slope answer different questions.

The time trend and Bitcoin interaction also limit a simple interpretation of a persistent event effect. ERCOT's log-total-moment time-slope change is now -0.616 per year with $p = 0.0011$, whereas its immediate reference-condition break is positive. That does not establish the direction of the entire post-period trend without the pre-period slope and the changing Bitcoin path. The point contrast at a later date is Equation 5, not the immediate coefficient alone.

9 Robustness, Pre-Event Dynamics, and Identification Limits

9.1 What the supplied sensitivity tests cover

The window, alternative-date, and winsorization checks concern daily mean basis, not the volatility models. Table 15 reports all primary immediate ERCOT/PJM coefficients from those checks. The corresponding Massena results are in Appendix B.1; none of its immediate mean coefficients is significant at 5 percent in those checks.

Table 15: Mean-basis sensitivity: primary hashprice specification

Comparison	Specification	$\widehat{\beta}$	95% CI	p
ERCOT: SOUTH	1-year window	69.390	[-18.332, 157.111]	0.1211
ERCOT: SOUTH	2-year window	3.453	[-3.158, 10.065]	0.3060
ERCOT: SOUTH	3-year baseline	4.451	[-1.989, 10.890]	0.1755
ERCOT: SOUTH	Date: 2021-05-26	2.330	[-6.029, 10.689]	0.5848
ERCOT: SOUTH	0.5% winsorized	3.999	[0.132, 7.865]	0.0427
PJM: candidate 1	1-year window	-19.914	[-47.662, 7.834]	0.1595
PJM: candidate 1	2-year window	2.264	[-1.111, 5.638]	0.1886
PJM: candidate 1	3-year baseline	2.659	[-0.209, 5.527]	0.0692
PJM: candidate 1	Date: 2023-04-01	1.926	[-0.304, 4.156]	0.0905
PJM: candidate 1	0.5% winsorized	2.391	[-0.061, 4.842]	0.0559
PJM: candidate 2	1-year window	-17.231	[-46.140, 11.679]	0.2427
PJM: candidate 2	2-year window	3.640	[-0.148, 7.428]	0.0597
PJM: candidate 2	3-year baseline	3.414	[0.186, 6.641]	0.0382
PJM: candidate 2	Date: 2023-04-01	2.710	[0.186, 5.234]	0.0353
PJM: candidate 2	0.5% winsorized	2.905	[0.300, 5.509]	0.0288

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice model, daily mean basis only. Windows are years on each side. Winsorization uses the original pre-period daily-basis 0.5th and 99.5th percentiles. Alternative-date regressions retain the original Bitcoin centering reference. These additional tests have no multiplicity adjustment and are not volatility robustness tests.

ERCOT's one-year mean coefficient is much larger than the longer-window estimates but

remains imprecise. Its winsorized estimate is nominally significant, while its untrimmed baseline and alternative-date results are not. Both PJM one-year coefficients are negative and imprecise. Candidate 2's alternative-date and winsorized estimates are nominally positive, while candidate 1's are not significant at 5 percent. These are unadjusted specification searches, not proof that the primary mean result is robust under every choice.

The alternative-date fits retain the original Bitcoin centering reference; their immediate coefficients are therefore not re-centered to the new date's Bitcoin condition. None of these checks establishes robustness of the log-volatility results. No volatility-window, storm-exclusion, alternative-offset, or full-first-stage-refit sensitivity estimates are included in this run.

9.2 Placebo dates and descriptive event bins

The restored export includes placebo output, which was absent from the earlier manuscript's evidence package. The placebo uses pre-event observations only and places a pseudo-event one year before the nominal event. ALCOA and St Lawrence have nominally positive placebo mean breaks, with $p = 0.0163$ and $p = 0.0366$, and nominally negative placebo slope changes. Those results are reasons for caution about attributing every fitted break to the facility event. The other placebo immediate coefficients are unresolved, which does not validate their counterfactuals.

Table 16: Pre-period pseudo-event mean-basis diagnostics

Comparison	Pseudo-date	$\widehat{\beta}$	95% CI	$p(\beta)$	$p(\kappa)$
Massena: SITHE	2017-05-01	-1.945	[-4.616, 0.726]	0.1535	0.2774
Massena: ALCOA	2017-05-01	0.377	[0.069, 0.684]	0.0163	0.0032
Massena: Cedars	2017-05-01	-1.691	[-4.484, 1.102]	0.2354	0.3809
Massena: St Lawrence	2017-05-01	0.411	[0.026, 0.797]	0.0366	0.0133
ERCOT: SOUTH	2019-05-01	6.341	[-2.633, 15.315]	0.1661	0.0617
PJM: candidate 1	2022-02-01	4.533	[-5.039, 14.104]	0.3533	0.9421
PJM: candidate 2	2022-02-01	2.397	[-9.108, 13.902]	0.6830	0.5974

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice specification fitted only before the actual nominal event. The pseudo-date is one year earlier; the original Bitcoin reference is retained. Values are dollars per MWh for the level coefficient. Both level and slope tests are exploratory and unadjusted.

The event-bin diagnostic groups observations in three-month bins, omitting the final pre-event quarter and pooling more distant observations into endpoint bins -8 and 7. It uses annual sine/cosine terms, weekdays, weather, gas, and the specified Bitcoin terms rather than the baseline linear-trend/month-indicator equation. In the primary version, ERCOT's bins -7 and -6 differ from the reference by approximately \$4.20 and \$4.43/MWh with raw p values of 0.0150 and 0.0193. Some Massena and PJM pre-event bins also differ individually. No joint lead test or untreated path is supplied. The manuscript therefore does not state that the analysis passed a pre-trend test.

9.3 Measurement, omitted conditions, and inference scope

The persistence of results after adding Bitcoin controls does not establish that all mining-market influences have been removed. Hashprice is a network revenue indicator, not metered utilization or a facility's complete operating margin. Population, generator outages, reserves, regional delivered fuel costs, detailed infrastructure events, and actual facility load are not in the baseline. Missing PJM weather remains a source of sample selection. The script also does not establish whether a particular extreme market episode drives the ERCOT findings.

Electrical relevance and event timing remain more fundamental than the quantity of downloaded prices. SOUTH cannot be called a verified Rockdale nodal effect, and the PJM candidates remain internal labels until their source identities and facility relevance are documented. A nominal post indicator is not an observed first-stage load increase. Price comparisons sharing a benchmark or event are not independent facility experiments, and removing exact aliases does not make them statistically independent.

The supplied controls and uncertainty calculations support conditional event-associated comparisons. They do not establish effects on retail bills, uniform responses per MW, congestion alone, or total long-run system costs. The opposite ERCOT and PJM signs have not been subjected to a formal cross-case equality test. Differences in individual significance are not themselves a test that the effects differ.

10 Discussion: Relevance to AI-Driven Data-Center Demand

The clearest finding is not a common immediate increase in average relative prices. It is the distinction between unresolved mean-basis changes and outcome-specific changes in price risk. The primary ERCOT log-total-moment result remains positive after restoring mining-market controls, while both PJM estimates remain negative and only the second passes the primary correction. Arithmetic residual comparisons continue to show declines at both PJM candidates. This combination illustrates why the outcome scale and comparison period must be stated rather than summarized as one general volatility effect.

These distinctions are relevant to the increase in data-center electricity demand associated with AI. Large computing loads are an electricity-system problem because of where, when, and under what operating arrangements power is required. A basis-oriented framework can reveal relative-price behavior that aggregate consumption or local mean prices alone cannot show. The mining cases supply evidence about that related problem, not an empirical observation of the same workloads now being deployed for AI.

The connection is strongest at the level of the research questions and measurement framework. Generalizing a numerical coefficient requires more: similar demand profiles, response during constrained hours, commissioning patterns, network positions, and supporting infrastructure. Those conditions are not established here. In particular, no assumption of equal or nearly full electrical utilization is imposed, and the estimates are not proposed as upper or lower bounds for AI effects.

The operator evidence discussed in Section 1 also cautions against treating flexibility as an absolute mining-versus-AI distinction.

Restoring Bitcoin controls makes the empirical analysis more faithful to mining economics, but it does not remove all workload-specific behavior to leave a generic data-center coefficient. Nor does AI relevance eliminate the earlier causal-identification problem. A stronger study would first establish what the mining event changed, then assess whether those conditions apply elsewhere. The current evidence supports an AI-adjacent interpretation of the electricity-market problem without relabeling the estimated treatment.

11 Conclusion

This paper examines public wholesale-price comparisons around nominal mining-related events in NYISO, ERCOT, and PJM, conditional on weather, gas, and lagged Bitcoin-market conditions. The primary hashprice-interaction model does not yield an immediate mean-basis coefficient that passes the reported 5-percent FDR adjustment. It does yield a 333.2-percent fitted geometric daily residual-moment increase in ERCOT. PJM's corresponding estimates are decreases of 36.7 and 47.0 percent, but only the second survives the primary adjustment. These results replace the earlier no-Bitcoin adjusted estimates rather than merely adding a new label to them.

On the arithmetic scale, below-fitted-mean changes account for 84.8 percent of ERCOT's total-moment increase and 95.8–96.9 percent of the PJM declines. Directional log coefficients, zero-day frequencies, and tail probabilities give additional, sometimes different information. The updated ERCOT upper-tail coefficient does not pass FDR, while St Lawrence's upper-tail increase does. The paper retains the imprecise and conflicting results alongside the stronger findings.

The significance of these mining cases for AI-driven demand is their connection to the broader integration of large computing loads, not proof that AI facilities have the same effects or run at the same utilization. Actual facility demand, electrical mappings, timing, sample selection, and overlapping market changes remain unresolved. The contribution is a Bitcoin-market-adjusted account of relative-price and distributional patterns, with a clear distinction between the empirical mining sample and its broader relevance to data-center growth.

A Raw Variance and Primary Secondary Trends

The raw hourly variances are retained separately from the residual-moment models. The following slope tables contain the primary hashprice coefficients, not the old no-Bitcoin slopes. A slope change can be present without a precise immediate mean break, and no slope or later-horizon test is included in the immediate-post FDR family.

Table 17: Unadjusted hourly basis variances and post/pre ratios

Comparison	Pre variance	Post variance	Ratio	95% block CI
Massena: SITHE	88.07	57.64	0.654	[0.229, 3.529]
Massena: ALCOA	2.54	9.13	3.599	[0.996, 9.817]
Massena: Cedars	90.33	92.19	1.021	[0.311, 4.865]
Massena: St Lawrence	4.77	9.61	2.016	[0.566, 7.536]
ERCOT: SOUTH	735.66	1,382.02	1.879	[0.641, 4.702]
PJM: candidate 1	352.06	89.28	0.254	[0.115, 0.432]
PJM: candidate 2	352.83	103.45	0.293	[0.125, 0.659]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Sample variance of paired hourly basis, in $(\$/\text{MWh})^2$. No weather/gas/Bitcoin eligibility screen. Ratios compare whole regimes; the null reference is one. Intervals are descriptive 499-draw, seven-day block percentile intervals.

Table 18: Primary changes in the daily mean-basis time trend

Pricing comparison	$\widehat{\kappa}$	95% CI	p
Massena: SITHE	0.322	[-0.049, 0.692]	0.0886
Massena: ALCOA	-0.060	[-0.187, 0.067]	0.3513
Massena: Cedars	0.385	[0.001, 0.770]	0.0494
Massena: St Lawrence	-0.089	[-0.226, 0.049]	0.2064
ERCOT: SOUTH	0.329	[-3.891, 4.549]	0.8786
PJM: candidate 1	0.471	[-1.140, 2.083]	0.5667
PJM: candidate 2	1.066	[-0.632, 2.764]	0.2187

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Units: dollars per MWh. Primary hashprice specification. Slope changes are per year, not the entire post-event trend. These secondary tests are not covered by the immediate-post FDR adjustment.

Table 19: Primary changes in the log-total-moment time trend

Pricing comparison	$\widehat{\kappa}$	95% CI	p
Massena: SITHE	0.241	[-0.115, 0.598]	0.1843
Massena: ALCOA	-0.371	[-0.670, -0.072]	0.0152
Massena: Cedars	0.052	[-0.297, 0.400]	0.7717
Massena: St Lawrence	-0.340	[-0.631, -0.049]	0.0221
ERCOT: SOUTH	-0.616	[-0.985, -0.246]	0.0011
PJM: candidate 1	-0.166	[-0.383, 0.050]	0.1324
PJM: candidate 2	-0.308	[-0.549, -0.068]	0.0121

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Units: log moment. Primary hashprice specification. Slope changes are per year, not the entire post-event trend. These secondary tests are not covered by the immediate-post FDR adjustment.

B Massena Directional Moments and Specification Comparisons

The Massena arithmetic decomposition uses the new primary hourly residuals. It is not a reprint of the former weather/gas residual table. All mean/control specifications are retained below, including the Bitcoin-price alternative where the St Lawrence total-moment coefficient is nominally significant. Such a result does not replace the primary hashprice specification.

Table 20: Massena: primary arithmetic residual-moment comparisons

Comparison	Moment	Pre level	Post level	Ratio	95% block CI
Massena: SITHE	Total M	87.81	57.74	0.658	[0.244, 4.965]
Massena: SITHE	Positive M^+	10.71	42.49	3.968	[1.694, 10.345]
Massena: SITHE	Negative M^-	77.11	15.25	0.198	[0.023, 21.741]
Massena: ALCOA	Total M	2.50	9.13	3.658	[0.942, 10.086]
Massena: ALCOA	Positive M^+	1.90	0.77	0.405	[0.177, 1.164]
Massena: ALCOA	Negative M^-	0.59	8.36	14.093	[3.073, 47.247]
Massena: Cedars	Total M	90.04	92.46	1.027	[0.289, 6.245]
Massena: Cedars	Positive M^+	12.78	46.46	3.635	[1.697, 7.800]
Massena: Cedars	Negative M^-	77.26	45.99	0.595	[0.036, 49.180]
Massena: St Lawrence	Total M	4.69	9.59	2.045	[0.542, 8.094]
Massena: St Lawrence	Positive M^+	1.35	0.85	0.627	[0.282, 3.024]
Massena: St Lawrence	Negative M^-	3.34	8.74	2.617	[0.588, 16.234]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice-adjusted hourly residuals. Levels are equal-weighted arithmetic means of usable daily moments, in $(\$/\text{MWh})^2$. All sign-specific moments use the same denominator. Seven-day, 499-draw percentile intervals condition on the fitted first stage and are descriptive.

B.1 Common-sample control comparisons

Table 21 retains the four common-sample control choices. Table 22 then reports the primary mean-basis sensitivity results; these are separate specification checks, not additional independent facilities.

Table 21: Massena: common-sample control specifications

Comparison	Controls	Mean $\widehat{\beta}$	Mean p	Moment (%)	Moment q_{42}
Massena: SITHE	Weather	0.573	0.2324	-17.3	0.6483
Massena: SITHE	Weather + gas	0.565	0.2372	-15.1	0.7132
Massena: SITHE	BTC-price interaction	1.054	0.0542	56.5	0.4163
Massena: SITHE	Hashprice interaction	0.943	0.0706	6.4	0.9055
Massena: ALCOA	Weather	-0.040	0.7231	-10.7	0.7212
Massena: ALCOA	Weather + gas	-0.047	0.6802	-11.2	0.7142
Massena: ALCOA	BTC-price interaction	-0.193	0.1363	29.8	0.5899
Massena: ALCOA	Hashprice interaction	-0.075	0.5318	1.3	0.9712
Massena: Cedars	Weather	0.503	0.3040	16.3	0.7203
Massena: Cedars	Weather + gas	0.494	0.3106	19.9	0.6987
Massena: Cedars	BTC-price interaction	1.063	0.0599	138.2	0.1214
Massena: Cedars	Hashprice interaction	0.967	0.0735	47.7	0.5320
Massena: St Lawrence	Weather	0.096	0.4130	43.2	0.3975
Massena: St Lawrence	Weather + gas	0.085	0.4589	37.2	0.4374
Massena: St Lawrence	BTC-price interaction	-0.005	0.9721	149.5	0.0513
Massena: St Lawrence	Hashprice interaction	0.098	0.4354	81.0	0.2244

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). All four specifications use the common eligible sample. Mean coefficients are dollars per MWh; moment entries are $100[\exp(\widehat{\beta}_M) - 1]$, not raw variance changes. q_{42} is recomputed within each specification. The original-sample weather/gas model agrees with its common-sample counterpart and is retained in the accompanying CSVs. Bitcoin-price and hashprice models are alternatives, not simultaneous regressors.

Table 22: Massena: primary mean-basis window and winsorization sensitivity

Comparison	Specification	$\widehat{\beta}$	95% CI	p
Massena: SITHE	1-year window	-5.191	[-17.581, 7.199]	0.4116
Massena: SITHE	2-year window	2.058	[-0.845, 4.961]	0.1647
Massena: SITHE	3-year baseline	0.943	[-0.079, 1.966]	0.0706
Massena: SITHE	0.5% winsorized	0.477	[-0.055, 1.009]	0.0786
Massena: ALCOA	1-year window	3.447	[-0.088, 6.982]	0.0560
Massena: ALCOA	2-year window	-0.246	[-0.714, 0.221]	0.3015
Massena: ALCOA	3-year baseline	-0.075	[-0.311, 0.160]	0.5318
Massena: ALCOA	0.5% winsorized	-0.069	[-0.263, 0.124]	0.4814
Massena: Cedars	1-year window	-3.378	[-16.303, 9.547]	0.6085
Massena: Cedars	2-year window	1.968	[-1.003, 4.938]	0.1942
Massena: Cedars	3-year baseline	0.967	[-0.092, 2.026]	0.0735
Massena: Cedars	0.5% winsorized	0.430	[-0.137, 0.996]	0.1369
Massena: St Lawrence	1-year window	0.705	[-2.668, 4.078]	0.6820
Massena: St Lawrence	2-year window	-0.236	[-0.695, 0.223]	0.3143
Massena: St Lawrence	3-year baseline	0.098	[-0.149, 0.346]	0.4354
Massena: St Lawrence	0.5% winsorized	0.089	[-0.114, 0.293]	0.3898

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Primary hashprice model, daily mean basis only. Windows are years on each side. Winsorization uses the original pre-period daily-basis 0.5th and 99.5th percentiles. Alternative-date regressions retain the original Bitcoin centering reference. These additional tests have no multiplicity adjustment and are not volatility robustness tests.

C Reference-Condition Horizon Contrasts

The supplied horizon export calculates $\beta + \kappa h$, with Bitcoin conditions fixed at the original event reference, for supported horizons. Table 23 shows the 12- and 36-month entries; all five horizons are retained in the numerical files. They are not actual-path predictions or average effects over a horizon. Their intervals use the joint coefficient covariance, but no multiple-testing correction is supplied for this set.

Table 23: Mean-basis point contrasts holding hashprice at its reference

Comparison	12 months	95% CI	36 months	95% CI
Massena: SITHE	1.265	[0.013, 2.517]	1.908	[0.056, 3.761]
Massena: ALCOA	-0.135	[-0.419, 0.148]	-0.256	[-0.732, 0.220]
Massena: Cedars	1.352	[0.071, 2.633]	2.123	[0.234, 4.012]
Massena: St Lawrence	0.010	[-0.286, 0.306]	-0.167	[-0.672, 0.337]
ERCOT: SOUTH	4.779	[0.094, 9.465]	5.437	[-4.038, 14.912]
PJM: candidate 1	3.130	[-0.868, 7.128]	4.073	[-2.789, 10.934]
PJM: candidate 2	4.479	[0.012, 8.947]	6.611	[-0.895, 14.118]

Source: supplied Bitcoin-control results (LMP Research Project, 2026e). Units: dollars per MWh. These are the existing exported reference-condition horizon contrasts, not new regressions. Positive or negative endpoint estimates do not establish the average post-period response or the response at a different Bitcoin condition.

D Replication Record and Remaining Documentation

The numerical basis is the supplied Bitcoin-control run `20260929T030053973839Z`, exported on September 29, 2026 at 03:34:29 UTC. The archive contains 21 compressed DataFrames and 84 coefficient covariance matrices. The source program is `LMP_Bitcoin_Controls_Spyder.py`; the electricity, weather, and gas checkpoints originate in `LMP_All_In_One_Spyder.py`. The previous `LMP_Analyze_Existing_Data.py` output is retained only through the explicitly identified no-Bitcoin comparison specification (LMP Research Project, 2026e,d,c,a). The UTC run date is distinct from the local manuscript revision date.

All 15 site–specification status rows report completion. The 470 saved design audits mark required controls present; those entries include auxiliary fits and are not 470 independent experiments. The five error-table entries are the expected exact-alias exclusion repeated across specifications. The primary mean and total-moment day counts are 2,182 per Massena comparison, 2,184 for ERCOT, and 2,003 per PJM candidate.

For this manuscript revision, the saved BH values were checked against the 42-test and 210-test families, all 84 covariance diagonals were checked against squared reported standard errors, and primary arithmetic moment levels were reconciled with the saved daily panel. Percentage transformations and directional shares use the unrounded CSV values. The post-hashprice sensitivities use the saved covariance matrices through Equation 6. These are consistency checks and algebraic

calculations, not an independent replication from original hourly inputs. No provider retrieval or regression was rerun to produce the revised manuscript.

The run manifest records the Bitcoin source snapshot hash and timestamp, but its executed-code hash is null. A local copy of the supplied script is preserved separately in the revision package; its hash does not retroactively establish the exact executed bytes. The full original Bitcoin source snapshot, original hourly input checkpoints, and an independently validated facility-to-node crosswalk are not supplied in the review archive. These limitations should remain explicit in a replication release.

The source IDs, exact source names, successful PJM query filters, station-history audit, backup-station share, software environment, and input hashes still need to be archived alongside the analysis. The supplied daily panel improves the ability to inspect dates and residual outcomes but does not by itself establish why weather is missing or how a facility operated. Neither a completed regression nor complete Bitcoin coverage verifies a load factor or a causal treatment.

Before stronger interpretation, the priorities remain electrical mapping and event documentation, common-sample raw/adjusted comparisons, alternative weather coverage, full-refit resampling, and volatility sensitivity to windows, extreme episodes, and offsets. The newly supplied placebo output and zero-day counts are now reported; a joint event-lead test and those additional volatility exercises remain uncompleted.

E Extension to Verified AI Loads

The connection to AI-driven data-center demand motivates an extension rather than a relabeling of the present sample. A future registry should distinguish announcement, construction, energization, first production load, and phased expansion. It should identify actual workload, net grid withdrawals, confidence in event timing, and overlapping generation or transmission changes. A commercial association with an AI company is not sufficient to classify an earlier mining event as AI deployment.

The same basis and distributional framework could then be applied with validated pricing points and credible untreated comparisons. Measured ramp-up and operating flexibility would permit a more direct evaluation of load intensity and constrained-hour behavior. Until those observations are established, the mining cases remain a related empirical setting that informs the electricity-market questions raised by AI growth without providing numerical AI-effect estimates.

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